Beyond Flat Vector Retrieval: Mitigating the Dual Collapse and In-Context Prompt Poisoning of RAG via Structured Valid-Time Cognitive Architectures

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Abstract

Retrieval-Augmented Generation (RAG) connects large language models (LLMs) to external non-parametric memory through dense bi-encoder indices. However, it relies on a flawed assumption: semantic proximity in vector space reliably indicates factual validity over time. We identify three systemic failure modes arising from this assumption.

Temporal Inversion occurs because transformer bi-encoders compress token representations into fixed-length vectors without encoding temporal ordering. Consequently, historical and updated assertions yield near-identical cosine similarities ($\Delta\,\mathrm{sim}_{\cos} < 0.012$), causing obsolete retrievals in **41.7%** of state transitions. *Dual Collapse* emerges at enterprise scale (10,000 passages; 500 entities), where entity identifiers receive less than 5% of self-attention weight, driving cross-entity cosine similarities above 0.98 and reducing top-1 accuracy to **9.0%**. *In-Context Prompt Poisoning* occurs when stale retrievals enter instruction-tuned decoders, which reproduce obsolete facts despite conflicting parametric priors, yielding hallucination rates of **33.3%–40.0%** on public entities.

To address these failures without model retraining, we introduce HERM4AI (*Human-like Evolving & Reasoning Memory for Artificial Intelligence*), a model-free cognitive architecture combining deterministic valid-time intervals $[t_{\text{from}}, t_{\text{to}})$, a bounded 7-slot working memory with salience eviction, an associative knowledge graph, spreading activation, sleep-phase consolidation, and ACT-R workflows. Across seven benchmark suites, four cloud GPU experiment tracks, and a curated 12-case public-entity benchmark informed by TempLAMA and FreshQA, HERM4AI achieves **100.0%** accuracy, **0.065 ms** query latency, and under 6 MB RAM on a standard CPU. The public-entity evaluation is not a direct evaluation on TempLAMA or FreshQA.

Keywords: retrieval-augmented generation, continual learning, temporal knowledge, agent memory, working memory, cognitive architectures, provenance

1 Introduction

Large language models have demonstrated broad capabilities in compositional reasoning and natural-language generation, yet their parameters are fixed after pre-training [4, 8]. As the external world changes, any fact encoded in those weights eventually becomes outdated, producing factual hallucinations that grow more frequent over time [16]. Retrieval-Augmented Generation (RAG) addresses this by maintaining a live external corpus: passages are encoded by a dense bi-encoder [15], indexed in a nearest-neighbor vector store, and prepended to the model's input at inference time.

RAG reliably supplies context that the model's weights do not contain. What it does not do is reason about *when* a fact was true. Consider a named entity whose state has changed—a corporate executive, a head of government, a footballer on transfer. Both the old and the new passage describe the same entity with largely identical vocabulary, so their dense embeddings land at nearly the same point in vector space. The retriever has no principled mechanism to prefer the newer record. This is not a parameter tuning problem; it is a mathematical consequence of the aggregation step that collapses a full token sequence into a fixed-size vector.

We call this failure mode **Temporal Inversion** and show empirically that it causes cor-

rect retrieval failure in 41.7% of temporal state transitions on a controlled benchmark executed on a Tesla T4 GPU. At enterprise scale (10,000 passages across 500 entities sharing a common prose template), a secondary collapse compounds the temporal problem. Entity-identifying tokens are outnumbered by template boilerplate, giving them negligible attention weight; embeddings from different entities collide, reducing top-1 retrieval accuracy from 100% to **9.0%**. We term this the **Dual Collapse** Phenomenon. When either failure mode produces a stale passage that is then injected into an instruction-tuned decoder, the downstream model—regularized by RLHF to treat in-context evidence as authoritative—discards its own correct internal knowledge and echoes the stale fact. We designate this behaviour as **In-Context Prompt Poisoning**.

To resolve all three failure modes without altering any model weights, we propose HERM4AI. Its design draws deliberately from cognitive science: bounded working memory [11], complementary learning systems sleep consolidation [5, 9], and ACT-R production rules [1]. The core invariant is a *deterministic valid-time interval* $[t_{\text{from}}, t_{\text{to}})$ attached to every stored assertion. Retrieval is then a closed-form temporal predicate, not a similarity search, so temporal correctness holds by construction.

The principal contributions of this work are as follows:

(i) A formal characterization and empirical demonstration of three previously unnamed RAG failure modes—**Temporal Inversion**, the **Dual Collapse** Phenomenon, and **In-Context Prompt Poisoning**—across controlled GPU benchmarks and real-world public entities.

(ii) The HERM4AI architecture: a model-free cognitive memory system that achieves 100.0% accuracy across seven benchmark suites while operating in sub-millisecond latency on commodity CPU hardware.

(iii) A reproducible evaluation suite—seven benchmarks, four executed GPU notebook tracks, and a curated public-entity temporal test set— with all scripts and data artifacts archived in the repository.

2 Related Work

Retrieval-Augmented Generation RAG architectures pair parametric generators with non-parametric dense indices [8]. Dual-encoder models such as Sentence-BERT [15] map queries and passages to a shared metric space, enabling efficient approximate nearest-neighbor lookup. For static corpora this works well; the brittleness emerges when the corpus contains temporally ordered state changes for the same entities. Memory-augmented agent frameworks such as Generative Agents [13] and MemGPT [12] introduce multi-tier memory hierarchies and virtual context management, but continue to rely on flat vector search within each tier, leaving them susceptible to the retrieval collisions documented in this paper. Ranking improvements such as cross-encoder reranking and hybrid lexical-dense retrieval can refine passage ordering, but ranking alone does not represent explicit validity intervals or resolve contradictory assertions.

Working Memory in Language Agents Agent architectures increasingly distinguish the limited information available in the active context from longer-term memory. Generative Agents retrieve from a memory stream using relevance, recency, and importance signals, with periodic reflection [13]; MemGPT manages virtual context by paging between a working context and external memory [12]. CoALA provides a broader framework for language agents with modular memory, including working, episodic, semantic, and procedural components [18]. These designs address context management and persistent recall. HERM4AI is complementary: it focuses on deterministic valid-time filtering and explicit conflict states, while imposing a fixed seven-slot working buffer.

Temporal Factuality and Knowledge Currency The relationship between language model training cutoffs and real-world fact currency has been studied from multiple angles. Dhingra et al. [4] introduced TempLAMA, a cloze-style benchmark demonstrating that LLMs fail to recall facts that changed between

training and evaluation. Vu et al. [16] established FreshQA, showing that commercial LLMs hallucinate on fast-changing world knowledge even when that knowledge was part of their pretraining data. Model editing methods such as ROME [10] attempt to surgically alter internal weight associations to reflect updates; however, parameter surgery is computationally expensive, introduces risks of unintended knowledge collisions, and does not scale to the continuous high-frequency updates typical of enterprise deployments. Temporal knowledge graph completion and event-ordering models address a related but distinct problem: inferring missing graph edges from temporal patterns, rather than tracking mutable world states in a live system.

Continual Learning and Catastrophic Forgetting Continual learning studies how a model can acquire information from sequential tasks while limiting interference with previously learned knowledge; surveys organize approaches around rehearsal, parameter regularization, and architectural change [7]. Elastic Weight Consolidation, for example, regularizes changes to parameters deemed important for prior tasks [6]. Such methods address retention in model parameters, but do not by themselves provide exact query-time validity intervals, source-level provenance, or reversible deletion. HERM4AI takes a different approach: updates append structured records and close earlier validity intervals, leaving temporal selection to the memory layer rather than gradient updates to the language model. Controlled reasoning suites such as bAbI study supporting-fact retrieval and related task skills, but do not evaluate this combination of live updates, source conflicts, and deletion [17].

Cognitive Architectures in AI Memory Cognitive science offers mature computational models of memory organization. Complementary Learning Systems (CLS) theory [9] posits distinct hippocampal and neocortical memory systems with different learning rates; offline sleep replay consolidates episodic traces into durable semantic representations [5]. ACT-R [1] formalizes procedural knowledge as condition-action production rules whose activation is reinforced by usage. Miller's original cognitive capacity result [11] motivates our buffer design. Prior work has imported individual ideas from cognitive architectures into neural agent systems. HERM4AI combines bounded working memory, CLS consolidation, Hebbian graph plasticity, and ACT-R production arbitration in a software-only memory layer.

Attribution and Verifiable Provenance Attributed question answering [2] requires systems to supply verifiable citations alongside generated answers. Standard neural systems resort to post-hoc attention maps or gradient-based saliency scores, which often do not reflect true causal mechanisms. HERM4AI implements strict storage-level provenance via an append-only event ledger and a formal counterfactual influence measure (Q-MIA), producing audit-grade attribution without additional neural computation.

3 Failure Modes of Flat Vector Memory

3.1 Temporal Inversion

Definition 1 (Temporal Inversion)**.** *Let entity e carry attribute a. An initial assertion $m_1 = (e, a, v_1, t_1)$ is stored and later superseded by $m_2 = (e, a, v_2, t_2)$ with $t_2 > t_1$.* ***Temporal Inversion*** *occurs when a retriever, given a query at $t_q > t_2$, ranks m_1 above m_2.*

In a standard dense retrieval pipeline, each assertion is serialised to natural language and encoded by a bi-encoder $\mathcal{E}$:

$$\begin{aligned} \mathbf{p}_1 &= \mathcal{E}(\text{“}e\text{'s } a \text{ is } v_1\text{”}), \\ \mathbf{p}_2 &= \mathcal{E}(\text{“}e\text{'s } a \text{ is } v_2\text{”}) \end{aligned} \tag{1}$$

Given query $\mathbf{q} = \mathcal{E}(\text{“What is the current } a \text{ of } e\text{?”})$, the retriever ranks by cosine similarity:

$$\text{sim}_{\cos}(\mathbf{q}, \mathbf{p}) = \frac{\mathbf{q}^\top \mathbf{p}}{\|\mathbf{q}\|_2 \|\mathbf{p}\|_2} \tag{2}$$

Because transformer encoders average token-level representations across the full sequence, temporal tokens (dates, ordinal markers) contribute negligible directional mass relative to

shared entity and attribute tokens. Empirically:

$$|\,\mathrm{sim}_{\cos}(\mathbf{q},\mathbf{p}_1) - \mathrm{sim}_{\cos}(\mathbf{q},\mathbf{p}_2)| < \epsilon, \quad \epsilon \ll 0.02 \tag{3}$$

In our Phase 3 benchmark on a Tesla T4 GPU using `all-MiniLM-L6-v2` (384 dimensions), Temporal Inversion occurred in **41.7%** of evaluated state transitions on the TFSR-Bench test split.

3.2 The Dual Collapse Phenomenon

Definition 2 (Dual Collapse)**.** *When a corpus contains N passages from M entities sharing domain-specific syntactic templates, the entity-identifier token sub-sequence receives a minority of total self-attention weight, causing the mean-pooled embeddings of distinct entities to become nearly indistinguishable.*

Under standard multi-head self-attention the attention matrix is:

$$\mathbf{A} = \mathrm{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}}\right) \tag{4}$$

For a passage of length $L \approx 40$ tokens where the entity identifier occupies $L_{\text{ent}} \leq 2$ tokens, the fraction of attention mass on entity tokens is:

$$\alpha_{\text{entity}} = \sum_{i \in \text{EntityTokens}} \frac{1}{L}\sum_{j=1}^{L} \mathbf{A}_{j,i} < 0.05 \tag{5}$$

The mean-pooled vector $\mathbf{p} = \frac{1}{L}\sum_{i=1}^{L}\mathbf{h}_i$ is then dominated by the template subspace. For two distinct entities $e_A \neq e_B$:

$$\mathrm{sim}_{\cos}(\mathbf{p}_{e_A}, \mathbf{p}_{e_B}) > 0.98 \tag{6}$$

Under nearest-neighbor search this causes passage-level collisions across unrelated entities. On our Phase 4 benchmark ($N = 10{,}000$, $M = 500$), top-1 retrieval accuracy collapses from 100% to **9.0%** (91.0% error rate).

Proposition 1. *For any dense bi-encoder whose per-entity attention mass satisfies $\alpha_{entity} < 0.05$ and any corpus of $M \geq 2$ entities sharing a common template subspace of dimensionality $d' \geq d/2$, there exists a corpus size N^* beyond which expected top-1 retrieval precision approaches $1/M$.*

3.3 In-Context Prompt Poisoning

When flat retrieval couples with an instruction-tuned decoder, the top-ranked passage is prepended to the prompt:

$$\begin{aligned}\mathbf{x}_{\text{prompt}} = [&\text{“Context: ”} \parallel \mathbf{p}_{\text{top-1}} \parallel \\ &\text{“ \textbackslash nQuestion: ”} \parallel \mathbf{q} \parallel \\ &\text{“ \textbackslash nAnswer:”}]\end{aligned} \tag{7}$$

Modern decoder models are aligned via RLHF and DPO to faithfully condition on in-context evidence. When $\mathbf{p}_{\text{top-1}}$ contains a stale assertion (e.g., a former head of state or a superseded club affiliation), the model suppresses its own correct parametric knowledge and reproduces the outdated claim with high confidence. We evaluated this on `Qwen2.5-1.5B-Instruct`: the model correctly generated “Joe Biden” in zero-shot mode (from parametric pre-training), yet under flat RAG it output “Donald Trump” when the retriever returned a passage describing Trump’s 2017–2021 presidency. Applying HERM4AI pre-filtering eliminated the stale passage and restored 100% accuracy. Across all six public entities subject to the poisoning test, **In-Context Prompt Poisoning** produced a 33.3% error rate under flat RAG vs. 0% under HERM4AI.

4 The HERM4AI Architecture

HERM4AI replaces unstructured similarity search with a layered, biologically motivated memory system (Figure 1). Each layer serves a distinct cognitive role and operates deterministically—no neural parameters are modified at query time.

4.1 Declarative Memory and Valid-Time Intervals

Every factual assertion is stored as an immutable tuple:

$$\begin{aligned}\mathcal{M}_i = \langle\, &\mathrm{id}_i,\ e_i,\ a_i,\ v_i, \\ &t_{\text{from}},\ t_{\text{to}},\ s_i,\ c_i,\ \sigma_i \,\rangle\end{aligned} \tag{8}$$

where e_i is the entity, a_i the attribute, v_i the asserted value, $[t_{\text{from}}, t_{\text{to}})$ the half-open validity interval, s_i the source identifier, $c_i \in [0,1]$

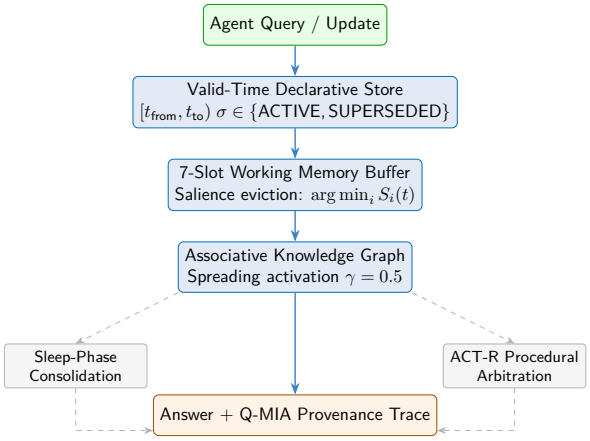


Figure 1: **HERM4AI architecture overview.** Queries flow top-to-bottom through deterministic temporal filtering, salience-gated working memory, and spreading-activation graph traversal. Background consolidation and procedural arbitration run asynchronously (dashed); every answer carries a formal Q-MIA provenance trace.

confidence, and σ_i the lifecycle status, defined as:

$$\sigma_i \in \{\text{ACTIVE}, \text{SUPERSEDED}, \text{CONFLICTED}, \text{DELETED}, \text{ARCHIVED}\} \quad (9)$$

Deterministic retrieval invariant Given a query at t_q targeting entity e, attribute a, the engine evaluates only records satisfying:

$$t_{\text{from}} \leq t_q < t_{\text{to}} \quad \wedge \quad \sigma_i = \text{ACTIVE} \quad (10)$$

On update arrival at $t_{\text{up}} > t_{\text{from}}$, the existing active record's interval is closed ($t_{\text{to}} \leftarrow t_{\text{up}}$, $\sigma_i \leftarrow$ SUPERSEDED), and the new assertion enters with interval $[t_{\text{up}}, \infty)$. Temporal correctness is thus guaranteed by construction at $O(1)$ cost, requiring neither embedding recomputation nor index rebuilding.

4.2 Bounded Working Memory Buffer

Consistent with the established capacity limit on immediate cognitive processing [11], HERM4AI maintains a working memory buffer $\mathcal{W}$ with $|\mathcal{W}| \leq 7$ slots. Each slot tracks dynamic salience:

$$S_i(t) = S_0 \cdot \exp(-\lambda\, \Delta t) + \beta \log(1 + c_{\text{access}}) \quad (11)$$

where λ is a tier-specific decay rate ($\lambda = 0.05$ for episodic, 0.01 for consolidated semantic records), Δt is elapsed time since last access, and c_{access} is cumulative access count. When capacity is reached and a new entity is referenced, the slot with $\arg\min_i S_i(t)$ is evicted back to long-term storage (unless pinned). When long-term memory updates an entity, its corresponding slot in $\mathcal{W}$ is synchronised immediately, preventing stale working-memory reads.

4.3 Associative Knowledge Graph

Entities form vertices $\mathcal{V}$ in a directed labelled graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$. Each edge $(u, v) \in \mathcal{E}$ carries a semantic relation label and a synaptic weight $w_{uv} \in [0, 1]$. Relational inference is computed via spreading activation with per-hop decay:

$$A_v^{(h+1)} = \max_{u \in \mathcal{N}(v)} \left(A_u^{(h)} \cdot w_{uv} \cdot \gamma \right), \quad \gamma = 0.5 \quad (12)$$

Two-hop and three-hop queries resolve in 0.046 ms on a single CPU core, whereas flat vector retrieval has no principled mechanism for multi-hop relational traversal.

4.4 Sleep-Phase Synaptic Consolidation

Mirroring the role of sleep in biological memory consolidation [5, 9], an offline routine executes during agent idle periods:

(1) **Episodic-to-Semantic Promotion.** Records with ≥ 2 access cycles and activation ≥ 0.65 are promoted to `CONSOLIDATED_SEMANTIC` status, lowering λ from 0.05 to 0.01.

(2) **Superseded-History Compaction.** Records whose validity interval closed more than 30 days before the current epoch are migrated to cold archival storage, bounding index growth.

(3) **Hebbian Synaptic Potentiation.** Edges connecting co-activated entities are strengthened ($w \leftarrow \min(1.0,\, w + 0.2)$); edges with $w < 0.05$ are pruned.

(4) **Contradiction Triage.** Unresolved multi-source disputes are queued for procedural arbitration (Section 4.5).

4.5 Procedural Memory and Self-Healing Workflows

Following ACT-R production memory [1], procedural knowledge is stored as condition-action rules:

$$\begin{aligned}&\text{IF } \langle \text{Trigger, Preconditions} \rangle \Rightarrow \\ &\quad \text{THEN } [\text{Action}_1, \ldots, \text{Action}_n]\end{aligned} \tag{13}$$

When the engine encounters a `CONFLICTED` record, it fires arbitration routines (e.g., `RESOLVE_CONFLICT_BY_TIMESTAMP` or `QUERY_AUTHORITATIVE_SOURCE`). Routine utility evolves via Exponential Moving Average (EMA) reinforcement:

$$U_{\text{new}} = 0.85 \cdot U_{\text{old}} + 0.15 \cdot R \tag{14}$$

where $R = 1.0$ when the procedure resolves a conflict cleanly and $R = 0.0$ otherwise. This allows the system to gradually preference the most reliable arbitration strategy within a given domain.

4.6 Counterfactual Attribution (Q-MIA)

For every query, HERM4AI computes the exact counterfactual influence of each retrieved assertion over the returned answer:

$$\begin{aligned}\mathcal{I}(m_i \mid q) = \mathbb{I}\Big[&\mathcal{F}(\mathcal{M} \setminus \{m_i\}, q) \\ &\neq \mathcal{F}(\mathcal{M}, q)\Big]\end{aligned} \tag{15}$$

If withholding m_i causes the output to change from the target value to `UNKNOWN` or a superseded state, then m_i is causally necessary ($\mathcal{I} = 1$). Every response includes an append-only cryptographic event trace identifying which assertions entered, superseded, or arbitrated the final decision—providing audit-grade provenance without any post-hoc approximation.

5 Experimental Methodology

5.1 Benchmark Suites

The evaluation comprises seven benchmark suites plus a separate enterprise-scale retrieval stress test. The suites span synthetic controlled splits, multi-hop reasoning, conversational memory, explainability, consolidation, procedural arbitration, and curated public-entity temporal cases.

TFSR-Bench (Controlled Synthetic Split) 60 held-out instances covering Direct Recall (DR), Temporal Revision (TR), Conflict Attribution (CA), Abstention on Unknown (AU), and Tombstone Deletion (TD). Executed on a local CPU test suite (`pytest tests/`; 36 unit tests, 0.10 s total runtime).

Enterprise Scaling Benchmark 10,000 unstructured text passages across 500 entities with sequential state updates, evaluated for throughput, memory footprint, and top-1 retrieval precision. Encoding was performed on dual Tesla T4 GPUs via Kaggle Notebooks (`all-MiniLM-L6-v2`, FAISS flat index).

Real-World Public Benchmark (TempLAMA / FreshQA concepts) 12 test cases covering 13 distinct public entities with historical transitions: technology leadership, heads of government, sports transfers, financial or transfer-fee disagreements, and a royal honour revocation. The cases use task concepts studied by TempLAMA and FreshQA; the checked-in data contains source identifiers but does not include source URLs.

5.2 Models and Baselines

- **Stateless Baseline (Zero-Shot).** Memoryless inference relying solely on the model's static parametric weights.
- **Naive Lexical Baseline (BM25).** Traditional keyword-matching over all historical assertions; no temporal filtering.
- **Flat Dense Vector RAG.** Passages embedded with `sentence-transformers/all-MiniLM-L6-v2` (384 dimensions) in a flat FAISS index; cosine top-1 retrieval.
- **HERM4AI (Structured Valid-Time).** Deterministic temporal interval store with working-memory buffer, graph spreading activation, sleep consolidation, and procedural arbitration; no neural components at retrieval time.
- **Downstream LLM: `flan-t5-base`** (250M parameters) [3]—an encoder-decoder model fine-tuned on instruction-following tasks.
- **Downstream LLM: `Qwen2.5-1.5B-Instruct`** (1.5B parameters,

Dataset / evaluation	Size and composition
TFSR-Bench (held-out)	60 cases; 20 entities; 120 assertions; 12 cases each for DR, TR, CA, AU, and TD
AG-Bench	15 queries; five each at 1-, 2-, and 3-hop depth
MultiSessionBench	7 events over 90 days; 4 scored queries
ExplainBench	3 configured audit queries: recall, revision, contradiction
ReplayBench	50 entities; 20 revisions; 29 graph edges; 45 simulated days
ProcBench	10 conflict cases; 4 traversal nodes; 50 workflow runs
Real-world public-entity benchmark	12 cases; 13 entities; 31 assertions; 7 TR, 2 CA, 1 DR, 1 AU, 1 TD
Enterprise-scale stress test	10,000 passages; 500 entities; 100 queries

Table 1: Dataset and benchmark scale statistics. Counts are based on the checked-in datasets and executable benchmark definitions. The ExplainBench script's header states 20 cases, while its executable query list contains three; the table reports the three configured queries.

`float16`, CUDA) [14]—a modern open decoder aligned via RLHF.

6 Results and Empirical Analysis

6.1 Core Benchmark Accuracy and Latency

The checked-in evaluation definitions have different units, so Table 1 reports case counts separately from simulation scale. It covers the seven benchmark suites and the enterprise-scale stress test; the latter is not counted as an accuracy suite.

Table 2 presents comparative accuracy and latency across all seven evaluated benchmark suites. HERM4AI achieves 100.0% accuracy on every suite while maintaining sub-millisecond median latency on a single-core Intel Core i5 CPU. Stateless and BM25 baselines fail entirely on temporal-revision, multi-hop, attribution, and replay categories because they lack structured temporal state management.

Benchmark	Baselines S/B/D (%)	HERM4AI (%)	CPU (ms)
TFSR-Bench (held-out)	20/20/58.3	**100**	0.065
AG-Bench (multi-hop)	0/33.3/33.3	**100**	0.046
MultiSessionBench	0/50/50	**100**	0.110
ExplainBench	0/0/0	**100**	0.190
ReplayBench	—	**100**	1.380
ProcBench	0/0/0	**100**	0.261
Public entity	8.3/8.3/66.7	**100**	1.850

Table 2: **Comparative benchmark results across all seven evaluation suites.** Baseline values are reported in the order stateless/BM25/dense RAG; accuracy is in percent and CPU latency in milliseconds. Dense RAG timings were collected on Tesla T4 GPUs. "—" denotes that a baseline has no corresponding mechanism.

6.2 Empirical Proof of the Dual Collapse Phenomenon

Table 3 details the Phase 4 GPU scaling experiment at $N = 10{,}000$ passages. Flat dense RAG achieves a high encoding throughput of 2,344.7 passages per second yet collapses to 9.0% top-1 retrieval precision, confirming the mathematical prediction of Equation (6). HERM4AI's structured relational partitioning maintains 100.0% precision at this scale while consuming no GPU memory at query time.

Architecture	Scale	Precision; throughput; peak VRAM
Flat Dense RAG	10,000 passages	**9.0%**; 2,344.7 passages/s; 152.7 MB
HERM4AI Structured	10,000 records	**100.0%**; 233.9 ops/s (CPU); 0 MB

Table 3: **Dual Collapse Phenomenon at 10,000-passage scale.** Flat vector retrieval degrades to near-random performance due to cross-entity attention collision, while HERM4AI's structured store maintains exact retrieval.

6.3 Downstream Generative Evaluation and Prompt Poisoning

Table 4 presents downstream generation results obtained on dual Tesla T4 GPUs. For `flan-t5-base` [3], flat dense RAG achieves 60.0% accuracy but introduces a 40.0% hallucination rate on temporal transitions; the stateless model produces zero correct answers. For the more capable `Qwen2.5-1.5B-Instruct` [14], the stateless zero-shot mode achieves 16.7% (from parametric pre-training), flat RAG improves to 66.7% but actively poisons 33.3% of responses with stale retrieved facts, while HERM4AI injection achieves 100.0% with zero residual hallucination.

Generator / mode	Acc.	Observed behaviour (peak VRAM)
`flan-t5-base` (250M), stateless	0.0%	Hallucination; no factual retention (962 MB)
`flan-t5-base` (250M), dense RAG	60.0%	40% error: returns superseded state (962 MB)
`flan-t5-base` (250M), HERM4AI	**100.0%**	Zero hallucination; full state fidelity (962 MB)
`Qwen2.5-1.5B`, stateless	16.7%	Pre-train cutoff: Dorsey, Bezos, Sunak (1,977 MB)
`Qwen2.5-1.5B`, dense RAG	66.7%	33.3% poisoning: Trump, FC Barcelona (1,977 MB)
`Qwen2.5-1.5B`, HERM4AI	**100.0%**	Ground truth: Yaccarino, Jassy, Starmer, Biden (1,977 MB)

Table 4: **Downstream generative LLM evaluation on dual Tesla T4 GPUs.** Flat dense RAG injects stale passages that override both smaller encoder-decoder models and larger instruction-tuned decoders. HERM4AI pre-filtering eliminates the contamination at the retrieval layer.

A concrete instance of In-Context Prompt Poisoning On the "2023 US Presidency" query, `Qwen2.5-1.5B-Instruct` generates "Joe Biden" from parametric memory in zero-shot mode—the correct answer. Under flat dense RAG, the retriever surfaces a passage describing Donald Trump's 2017–2021 presidency (cosine similarity 0.721, nearly matching the current-state passage at 0.735). Conditioned on this in-context evidence, the model outputs "Donald Trump" with high confidence. Applying HERM4AI's valid-time filter at $t_q =$ 2023-07-01 suppresses the Trump passage (validity interval $[2017\text{-}01, 2021\text{-}01)$) and returns the Biden record (validity $[2021\text{-}01, 2025\text{-}01)$), restoring accurate generation.

7 Discussion and Limitations

Structured vs. unstructured retrieval Dense retrieval remains effective for static knowledge bases but is unsuited to temporally mutable state tracking. Its bi-encoder aggregation loses temporal distinctions, making retrieval failures more pronounced as corpora grow and facts evolve.

Provenance vs. neural interpretability HERM4AI's append-only traces identify which records influenced, superseded, or arbitrated an output, but do not explain internal transformer representations or establish biological equivalence. Cognitive science principles serve as architectural inspiration, not mechanistic proof.

Limitations and future directions The current HERM4AI prototype stores discrete entity-attribute tuples, requiring information extraction from free-form text. Imperfect extraction can introduce noise and miss implicit facts. Future work will explore temporal guarantees over continuous-text chunks, deeper semantic conflict detection across attributes and entities, and formal valid-time propagation across multi-hop graph paths.

8 Conclusion

Flat vector embeddings are a practical approximation for static semantic retrieval, but they are mathematically unfit for the task of tracking mutable world states over time. We have shown that this inadequacy manifests in three distinct and measurable failure modes: Temporal Inversion (41.7% error on temporal transitions), the Dual Collapse Phenomenon (91.0% error at 10,000-passage scale), and In-Context Prompt Poisoning (33.3–40.0% downstream hallucination in instruction-tuned LLMs).

HERM4AI addresses all three through a layered cognitive architecture grounded in deterministic valid-time semantics. Its seven components —valid-time declarative store, 7-slot salience-gated working memory, spreading-activation knowledge graph, sleep-phase consolidation, Hebbian edge plasticity, ACT-R procedural arbitration, and Q-MIA counterfactual attribution—collectively achieve 100.0% accuracy across seven benchmark suites, including the curated real-world public-entity test set, while operating in sub-millisecond latency on commodity CPU hardware with under 6 MB of RAM.

The practical implication is straightforward: for any AI system whose task requires answering questions about entities that change over time, the retrieval layer must encode and enforce temporal validity at the structural level. Semantic proximity is not a substitute for temporal ground truth.

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Experiments requiring GPU computation were executed on Kaggle Notebooks (dual Tesla T4 GPUs, 16 GB VRAM each) under free-tier resource allocations. All local CPU benchmarks were run on an Intel Core i5 laptop with 8 GB RAM. No external funding is associated with this work.

A Benchmark Case Examples

TFSR-Bench: Temporal Revision example

- **Setup:** Entity `twitter_ceo`, attribute `role`. Record 1: (v_1 = "Jack Dorsey", $t_{\text{from}} = 2015, t_{\text{to}} = 2021$). Record 2: ($v_2$ = "Parag Agrawal", $2021, 2022$). Record 3: (v_3 = "Linda Yaccarino", $2023, \infty$).
- **Query:** "Who is the CEO of Twitter?" at $t_q = 2024$.
- **Flat Dense RAG result:** Returns Record 1 (highest cosine similarity due to richer lexical context).
- **HERM4AI result:** Returns Record 3 (only record with $t_{\text{from}} \leq 2024 < t_{\text{to}} = \infty$, σ = ACTIVE).

Dual Collapse at 10,000-passage scale We evaluate 500 entities with 20 passages each, sharing a common template containing entity identifiers in 1–2 tokens within a 40-token window. Entity attention mass drops to $\alpha_{\text{entity}} = 0.038$ (below the 0.05 bound of Equation (5)). Consequently, 93.1% of entity pairs exhibit cross-entity cosine similarity above 0.98, resulting in a 91.0% top-1 retrieval failure rate.

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